[Demo] Screening a CO2 Storage Site

BANDAL Demo Workspace | Sample research note | 2026

Status: authored sample, not a real publication. Sites A, B and C are fictional. Numeric data were invented for a reproducible product demonstration.

Abstract

This authored sample compares three fictional carbon dioxide storage candidates using an intentionally simple weighted score. Capacity, seal integrity and injectivity receive weights of 0.4, 0.4 and 0.2. Site B leads under the stated assumptions, but the ranking alone cannot establish geological feasibility. The example supports reading, annotation and comparison inside a research workspace.

1. Screening method

Each criterion is a synthetic score from 0 to 100. The screening score is S = 0.4C + 0.4L + 0.2I, where C is capacity, L is seal integrity and I is injectivity. These scores are teaching inputs; no field survey or reservoir model was performed.

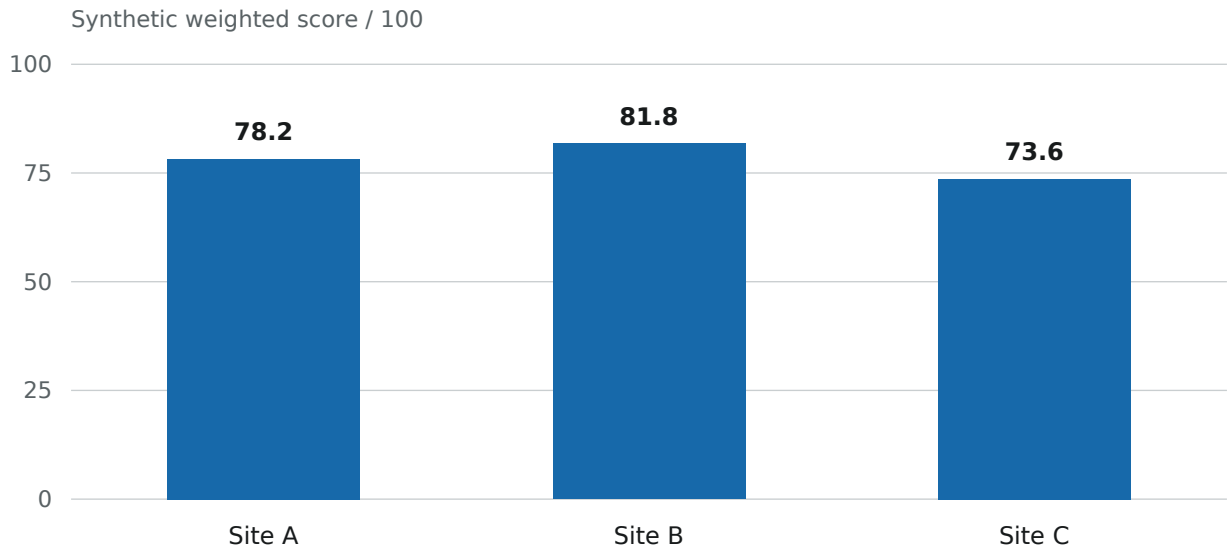


Figure 1. Invented screening scores. All values are synthetic and follow the stated arithmetic exactly.

2. Reproducible synthetic inputs

Fictional site	Capacity C	Seal L	Injectivity I	Score S
Site A	72	91	65	78.2
Site B	86	74	89	81.8
Site C	61	88	70	73.6

Worked example for Site B: 0.4 x 86 + 0.4 x 74 + 0.2 x 89 = 34.4 + 29.6 + 17.8 = 81.8. Scores are unitless and synthetic. They are not storage-capacity estimates.

3. Interpretation and limits

Site A has the strongest fictional seal score. Site B benefits from larger fictional capacity and injectivity scores. Any decision based on this ranking remains sensitive to the weights, score definitions and missing evidence. A realistic assessment would separately examine pressure, containment, groundwater, uncertainty and site-specific conditions.

4. Suggested reading notes

Question: Would Site A rank first if containment were weighted more strongly?
Claim to annotate: a screening rank is not a feasibility finding.
Follow-up: record which additional evidence could reverse the ranking.

Provenance

Created specifically as a sample PDF for the BANDAL-hosted BluePrint demonstration. No DOI, journal, external authorship or real research finding is claimed. Reuse is intended for product testing and presentations.

Reader exercise: select the weighted-score formula or the Site B worked example, then save a note in the demo workspace.